# A Extracellular electrophysiology (ecep

```
microephys_toy/
├── dataset_description.json
├── participants.tsv
├── probes/
│   └── customprobe1.json
├── sub-mouse01/
│   ├── ecephys/
│   │   ├── sub-mouse01_probes.tsv
│   │   ├── sub-mouse01_probes.json
│   │   ├── sub-mouse01_electrodes.tsv
│   │   ├── sub-mouse01_channels.tsv
│   │   ├── sub-mouse01_space-AllenCC
│   │   └── sub-mouse01_space-AllenCC
│   └── ses-01/ecephys/
│       ├── …_task-reach_ecephys.js
│       ├── …_task-reach_ecephys.nw
│       ├── …_task-reach_events.tsv
│       ├── …_task-rest_ecephys.jso
│       └── …_task-rest_ecephys.nwb
```

*_

| pro |
| --- |
| p |
| p |

## *_ecephys.json

# ɔhys)

## _probes.tsv

| ɔbe_name | type | manufacturer | model |
|---|---|---|---|
| probe01 | silicon-probe | NeuroNexus | A1x32-Poly3-10mm-50 |
| probe02 | silicon-probe | Custom Lab | customprobe1 |

**v**

Fv3_electrodes.tsv
Fv3_coordsystem.json

**son**
**wb/nix**
v
on
b

## *_electrodes.tsv

| name | probe_name | x |
|---|---|---|
| e001 | probe01 | 0 |
| e002 | probe01 | 0 |
| ⋮ | | |
| e033 | probe02 | 0 |

## *_channels.tsv

| name | electrode_name | type | units |
|---|---|---|---|
| ch001 | e032 | LFP | uV |
| ch017 | e032 | HP | uV |
| ch033 | e048 | LFP | uV |
| sync01 | n/a | SYNC | V |

*Physical recording devices*

| | electrode_count | hemisphere | associated_brain_region |
|---|---|---|---|
| )-177 | 32 | L | Primary Motor Cortex |
| | 16 | R | Hippocampus CA1 |

*Probe-relative contact positions*

| x | y | z | impedance | shank_id | material | location |
|---|---|---|---|---|---|---|
| ) | 0 | 0 | 1.2 | 0 | iridium-oxide | MOp |
| ) | 0 | 25 | 1.1 | 0 | iridium-oxide | MOp |
| | | | | | | |
| ) | 0 | 0 | 1.5 | 0 | iridium-oxide | CA1 |

*Recording channel properties*

| | sampling_frequency | gain | low_cutoff | high_cutoff | status |
|---|---|---|---|---|---|
| | 1000 | 500 | 0.1 | 300 | good |
| | 30000 | 500 | 300 | 10000 | good |
| | 1000 | 500 | 0.1 | 300 | good |
| | 30000 | 1 | n/a | n/a | good |

```
{ "TaskName": "Reach to Grasp",
  "SamplingFrequency": 30000,
  "Manufacturer": "IMEC",
  "SampleEnvironment": "in vivo", … }
```

# B Intracellular electrophysiology (icepl

```
└── sub-mouse02/
    ├── sub-mouse02_scans.tsv
    └── icephys/
        ├── sub-mouse02_probes.tsv
        ├── sub-mouse02_electrodes.
        ├── sub-mouse02_channels.tsv
        ├── …_task-IVcurve_icephys.
        ├── …_task-IVcurve_icephys.
        └── …_task-IVcurve_events.t
```

## *_icephys.json

```
{ "TaskName": "IV Curve
Characterization",
  "SamplingFrequency": 20000,
```

## *_coordsystem.json

```
{ "MicroephysCoordinateSystem": "All
  "MicroephysCoordinateUnits": "um",
  "MicroephysCoordinateSystemDesc
   "Origin at anterior commissure…" }
```

ıys)

### *_probes.tsv

| probe_name | type | manufacturer | elec |
|---|---|---|---|
| pipette01 | patch-pipette | Sutter | |
| pipette02 | patch-pipette | Sutter | |

tsv
v
json
nwb
sv

### *_electrodes.tsv

| name | probe_name | pipette_solution | internal_p |
|---|---|---|---|
| patch01 | pipette01 | K-gluconate | |
| patch02 | pipette02 | K-gluconate | |

### *_channels.tsv

| name | type | units | sampling_ |
|---|---|---|---|

lenCCFv3",

ription":

*Pipette devices*

| trode_count | hemisphere | associated_brain_region |
|---|---|---|
| 1 | L | Visual Cortex Layer 2/3 |
| 1 | R | Visual Cortex Layer 2/3 |

*Pipette-specific properties*

| pipette_diameter | external_pipette_diameter | material |
|---|---|---|
| 1.5 | 2.5 | borosilicate-glass |
| 1.5 | 2.5 | borosilicate-glass |

*Recording channels with mode*

| _frequency | recording_mode | gain | ground | status |
|---|---|---|---|---|

"Manufacturer": "Molecular Devices",
"SampleEnvironment": "ex vivo", … }

***Bold = REQUIRED***

| | | | |
|---|---|---|---|
| patch01_vm | VM | mV | 20 |
| patch01_im | IM | pA | 20 |
| patch02_vm | VM | mV | 20 |
| patch02_im | IM | pA | 20 |

| | | | | |
|---|---|---|---|---|
| )000 | current-clamp | 10 | AgCl | good |
| )000 | current-clamp | 5 | AgCl | good |
| )000 | voltage-clamp | 10 | AgCl | good |
| )000 | voltage-clamp | 5 | AgCl | good |